

ADVANCED SURVIVAL ANALYSIS

Multivariate Survival Analysis: Bayesian Survival Analysis. Frailty models.

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Introduction to Survival analysis from a Bayesian perspective. Main features.

Nowadays, Bayesian statistics are a truly widespread tool. The flexibility and the intuitive tools provided by Bayesian analysis make it an attractive option when dealing with statistical problems. First and foremost the reader should have in mind the Bayesian framework. In Bayesian analysis we have an statistical model $M = \{p(x|\theta), \theta \in \Omega\}$, which assumes that the observed data comes from a parametric model. Until now, we have the same components than in the frequentist approach. However, in Bayesian analysis we consider the parameter θ as a random variable and is willing to choose a probability distribution over Ω , $\pi(\theta)$, before looking at the data, that captures prior knowledge about θ . The distribution $\pi(\theta)$ is the so-called *prior distribution*. Then a Bayesian model has the following shape

$$M_B = \{p(x|\theta), \theta \in \Omega, \pi(\theta)\}. \quad (1)$$

Having the Bayesian model (1) we compute the so-called *posterior distribution*,

$$\pi(\theta|x) = \frac{p(x|\theta)\pi(\theta)}{p(x)} \propto p(x|\theta)\pi(\theta). \quad (2)$$

As the reader can easily see in (2), the posterior distribution is a compromise between the likelihood (the information given by the observed data) and the prior (the prior knowledge on the problem). Once we have the posterior distribution, we have all we need to do all the inference we want.

The above is the Bayesian framework. Now, it is time to comment on the effects of this framework when dealing with a survival analysis problem. The first topic to be commented

is the parametric modeling. Parametric models play an important role in Bayesian survival analysis since many Bayesian analyses in practice are carried out using parametric models. Parametric modeling offers straightforward modeling and analysis techniques. Due the construction of the Bayesian framework, the easiest way to work with a Bayesian model is under a clear parametric model. Nonetheless, in the case of the frequentist survival analysis, semi-parametric model is truly widespread. In fact, the most popular survival model, the Cox model, is a semi-parametric model. Modeling under a semi-parametric approach, or working with a non-completely defined parametric model, will add some difficulties when doing survival analysis under the Bayesian approach. Fortunately, there are plenty of references in the literature dealing with this problem and other that arises when using Bayesian analysis with survival problems. Here we will use [4] and [5] to address to the theoretical stuff.

The Cox model

Let T be the random variable time to the event ϵ . Remember that the Cox model expression is

$$\lambda(t|\mathbf{X}) = \lambda_0(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p), \quad (3)$$

where $\mathbf{X} = (X_1, \dots, X_p)'$ are p covariates, and $\lambda_0(t)$ is the baseline hazard function corresponding to $\mathbf{X} = \mathbf{0}$. Moreover, remember that $\lambda_0(t)$ contains the constant term in this model and can take any shape. This is why the model is called a *semiparametric* model. We know that in the frequentist approach the β 's are estimated via the partial likelihood

$$L_1(\beta_1, \dots, \beta_p) = \prod_{j=1}^r \frac{\exp(\beta' x_{(j)})}{\sum_{l \in R(t_{(j)})} \exp(\beta' x_l)}, \quad (4)$$

where $t_{(1)}, \dots, t_{(r)}$ are the r ordered times to the event of interest, $X_{(j)}$ the covariate vector for the individual failing at $t_{(j)}$, and $R(t_{(j)}) = \{i : Y_i \geq t_{(j)}\}$ is the set of individuals at risk at $t_{(j)}$. It can be proved (see the Cox's seminal paper [3]) that the β 's estimated via the maximization of (4) have great asymptotic properties. The estimate is consistent and asymptotically normal.

But, what happens when we use the Cox model within the Bayesian framework? In Bayesian analysis, the typical asymptotic results, like the ones commented above, do not apply. In addition, in Bayesian analysis one have to work always with the complete likelihood. Does not make sense in Bayesian analysis to use a partial likelihood or a pseudo-likelihood, the usage of non-complete likelihoods implies the lost of the observed data information in the Bayesian context. Hence, we should study how to calculate the complete likelihood under the Cox model.

The likelihood for right-censored survival data, based on any survival function model $S(t|x; \theta)$, can use two approaches. Let θ be the set of all model parameters and x is a vector of covariates. Let $f(t|x; \theta)$ be the density and $S(t|x; \theta)$ be the survival function. The likelihood of continuous data based on the right-censored survival data $D = (n, y, \nu, X)$ is:

$$L(\theta|D) = \prod_{i=1}^n f(y_i|x_i; \theta)^{\delta_i} S(y_i|x_i; \theta)^{1-\delta_i}, \quad (5)$$

where $y = (y_1, y_2, \dots, y_n)'$, $\delta = (\delta_1, \delta_2, \dots, \delta_n)'$ δ_i is the censoring indicator, y_i is the observed survival time subject to right-censoring, X is the $n \times p$ matrix of covariates with the i th row $\mathbf{x}_i'$.

When the survival model follows the Cox model (3), the likelihood of (5) can be expressed as

$$L(\theta|D) = \prod_{i=1}^n \exp\{-H_{0(i)} \sum_{j \in R_{(i)}} \exp(x'_j \beta)\} \{dH_0(y_{(i)}) \exp(x'_{(i)} \beta)\}^{\delta_{(i)}}, \quad (6)$$

where $\theta = (\beta, H_0(\cdot))$, $H_0(t) = \int_0^t \lambda_0(u) du$ is the baseline cumulative hazard function, $0 = y_{(0)} < y_{(1)} < \dots < y_{(n)}$ are the ordered observed survival times, $x_{(i)}$ is the covariate associated with $y_{(i)}$, $H_{0(i)} = H_0(y_{(i)}) - H_0(y_{(i-1)})$ is the increment in cumulative hazard between consecutive observed survival times $y_{(i-1)}$ and $y_{(i)}$, and $R_{(i)} = \{j : y_{(j)} \geq y_{(i)}\}$.

In practice, survival data is not observed in a continuous way. It is, we can assume that survival times are observed within intervals of time. Hence, the observed survival time y_i of any subject is only known up to an interval (however small the interval may be) where the length of the interval Δ_i depends. Let D_G denote the observed data, where we only know that the subject i has failed within an interval $(y_i, y_i + \Delta_i)$ when $\delta_i = 1$ and (y_i, ∞) when $\delta_i = 0$. To construct the likelihood corresponding to this observed discrete-time (or grouped) data D_G we first use a partition of the time, $0 < a_1 < a_2 < \dots < a_J < \infty$, where $a_J \geq$ maximum monitoring time, each $y_i = a_k$ and $y_i + \Delta_i = a_{k+1}$ for some k . The assumption about monitoring of subjects is that the grid interval is equal to the monitoring interval of each subject. For this situation, the likelihood is

$$L(\theta|D_G) = \prod_{i=1}^n \{S(y_i|\theta) - S(y_i + \Delta_i|\theta)\}^{\delta_i} S(y_i|\theta)^{(1-\delta_i)} = \prod_{j=1}^J G_j, \quad (7)$$

where $G_j = \exp[-\lambda_j \sum_{k \in \mathbb{R}_j - \mathcal{D}_j} \exp(x'_k \beta)] \{1 - \exp[-\lambda_j \sum_{k \in \mathcal{D}_j} \exp(x'_k \beta)]\}$, the risk-set $\mathcal{R}_k$ at time a_k is the set of people "at risk" in the interval $I_k = (a_{k-1}, a_k]$, $\mathcal{D}_k$ is the set of people

failing at that interval I_k and $\lambda_j = H_0(a_j) - H_0(a_{j-1})$. This likelihood of (7) has more fidelity to the actual observed data.

In the literature one can find that not much is gained in using a strictly nonparametric $\lambda_0(t)$ along with the discrete-time likelihood of (7). In practice, the implementation of the Bayesian inference with nonparametric $\lambda_0(t)$ is much more complicated than that of using a discretized parametric $\lambda_0(t)$. Thus, in practice, a discrete approximation of $\lambda_0(t)$ is used with the discrete-time likelihood.

The estimation of the regression coefficients in the Cox model under the frequentist approach can be obtained without specifying a model for the baseline hazard function by means of (4). This is not the case of Bayesian analysis which in general needs to specify some model for the baseline hazard. Baseline hazard functions can be modeled considering some of the usual probability distributions in the survival analysis framework such as exponential, Weibull, Gompertz, etc. However, these distributions are not as flexible as we would like and do not allow irregular behaviours. In order to work with a more flexible baseline hazard function the piecewise constant hazard model is a good option.

The piecewise constant hazard model is one of the most commonly used models. To construct this model, we first build a finite partition of the time axis, $0 < s_1 < s_2 < \dots < s_J$, with $s_J > y_i$ for all $i = 1, 2, \dots, n$. Thus, we have the J intervals $(0, s_1], (s_1, s_2], \dots, (s_{J-1}, s_J]$. In the j th interval, we assume a constant baseline hazard $\lambda_0(y) = \lambda_j$ for $y \in I_j = (s_{j-1}, s_j]$. Writing $\boldsymbol{\lambda} = (\lambda_1, \lambda_2, \dots, \lambda_J)'$, we can write the likelihood function of $(\boldsymbol{\beta}, \boldsymbol{\lambda})$ for the n subjects as follows

$$L(\boldsymbol{\beta}, \boldsymbol{\lambda} \mid D) = \prod_{i=1}^n \prod_{j=1}^J (\lambda_j \exp(\mathbf{x}'_i \boldsymbol{\beta}))^{\delta_{ij} \nu_i} \exp \left\{ -\delta_{ij} [\lambda_j (y_i - s_{j-1}) + \sum_{g=1}^{j-1} \lambda_g (s_g - s_{g-1})] \exp(\mathbf{x}'_i \boldsymbol{\beta}) \right\}, \quad (8)$$

where $\delta_{ij} = 1$ if the i th subject failed or was censored in the j th interval, and 0 otherwise. The indicator δ_{ij} is needed to properly define the likelihood over the J intervals. The semiparametric model in (8), sometimes referred to as a *piecewise exponential model*, is quite general and flexible to permit several shapes of the baseline hazard over the intervals. Moreover, we note that if $J = 1$, the model reduces to a parametric exponential model with failure rate parameter $\lambda = \lambda_1$. The piecewise constant hazard model is a useful and simple model for modeling survival data.

A common prior of the baseline hazard $\boldsymbol{\lambda}$ is the independent gamma prior $\lambda_j \sim \text{Gamma}(\alpha_{0j}, \lambda_{0j})$ for $j = 1, 2, \dots, J$. Here α_{0j} and λ_{0j} are prior parameters which can be elicited through the

prior mean and variance of λ_j .

The likelihood in (8) is based on continuous survival data. The likelihood function based on grouped/discretized survival data D_G of (7) for the piecewise constant hazard model is given by $L(\boldsymbol{\beta}, \boldsymbol{\lambda} \mid D_G) \propto \prod_{j=1}^J G_j$, where

$$G_j = \exp \left\{ -\lambda_j \Delta_j \sum_{k \in \mathcal{R}_j - \mathcal{D}_j} \exp(\mathbf{x}'_k \boldsymbol{\beta}) \right\} \prod_{l \in \mathcal{D}_j} [1 - \exp \{ -\lambda_j \Delta_j \exp(\mathbf{x}'_l \boldsymbol{\beta}) \}],$$

and where $\Delta_j = a_j - a_{j-1}$, $\mathcal{R}_j$ is the set of patients at risk, and $\mathcal{D}_j$ is the set of patients having failures in the interval I_j .

Frailty models

We start fixing some notation and presenting the model. Let T be the time until the event of interest ϵ . Assume that the sample units are clusters (e.g. patients are clustered within hospitals, patients itself could be a cluster, etc). Suppose we have n observations, p covariates and l (possibly $l = n$) clusters. Then the model expression for the so-called *Frailty model* (or shared frailty model) is

$$\lambda(t \mid \mathbf{X}, \text{Cluster } k) = w_k \lambda_0(t) \exp(\beta_1 X_1 + \cdots + \beta_p X_p), \quad (9)$$

where w_k , $k = 1 \dots, l$ is the frailty term corresponding to the cluster k , $\lambda_0(t)$ is the (unspecified) baseline hazard function, $\mathbf{X} = (X_1, \dots, X_p)'$ are the p covariates of the model. In (9) the reader can observe that the frailty term acts multiplicatively on the hazard. Moreover, usually W (the random variable of the frailty terms) is assumed to have mean 1.

In the Bayesian analysis framework we should study how to model the frailties, and also how to model the baseline hazard. As we commented before, in Bayesian analysis we can not don't specify the baseline hazard function as we do in the frequentist approach. For modeling the baseline hazard function it can be used the piecewise-constant baseline hazard. Nonetheless, in practice, many times a parametric model is assumed and then the baseline hazard function is specified by this parametric model.

Now, let us work on modeling the frailties. The frailty model (9) is completely specified by taking a distribution of the frailty W across the clusters. It is, a random-effect distribution must be specified. The most common frailty distribution used in the Bayesian literature is the gamma distribution. We assume that $W \sim \text{Gamma}(\kappa^{-1}, \kappa^{-1})$, and then we have that $E(W) = 1, \text{Var}(W) = \kappa$, so the frailty has mean 1 as we want. In addition, by using this approach we only have one parameter κ to estimate. Having a unit mean for the frailty is appropriate because then the frailty distribution serves only in the relative

sense of proportional variation across clusters, and the baseline hazard the magnitude of the average population. The gamma frailty offers a simple characterization of within-cluster dependence in terms of Kendall's tau being $1/(1 + 2/\kappa)$. Using the gamma frailty model the unconditional $(T, \mathbf{X})$ association does not preserve the proportionality of hazards. And so, we are interested on the conditional association $(T|W, \mathbf{X})$, in which the proportionality holds. One of the advantages of the gamma frailty model is the conditional conjugacy. Let $\mathbf{w}_{-k} = (w_1, \dots, w_{k-1}, w_{k+1}, \dots, w_l)$, then we have

$$(w_k | \mathbf{w}_{-k}, \boldsymbol{\beta}, \boldsymbol{\lambda}) \sim \text{Gamma}(\kappa^{-1} + r_k, \kappa^{-1} + g(\boldsymbol{\beta}, \boldsymbol{\lambda}, y_i)),$$

where r_k is the number of failures in the k-th cluster, and

$$g(\boldsymbol{\beta}, \boldsymbol{\lambda}, y_i) = \sum_{j=1}^l \exp(\boldsymbol{\beta}' x_{ij}) H_0(t_{ij}).$$

This property is very useful when using MCMC (Markov Chain Monte Carlo methods) to do the computations. In practice, we have to define the prior distribution of all the parameters playing a role in our model. A common strategy is to define less informative priors, there are some suggestions about this in the literature that will be used before to develop an example.

Case study

In this final section we deal with a case study and we see how the above concepts are applied in practice. The reference used for this case study section is [1]. The software used is R and Stan¹. Stan is a Bayesian software that allow us to work with Bayesian models in an easy manner. This software computes the posterior distributions by means of the MCMC method, specifically using the Hamiltonian Monte Carlo (HMC) method, which is a particular case of a MCMC (for further details on these topics see [2]).

In this case study we use the kidney infection data set, available in the *frailtyHL* package in R. This dataset consists of times to the first and second recurrences of infection in 38 kidney patients using a portable dialysis machine. Infections can occur at the location of insertion of the catheter. The catheter is later removed if infection occurs and can be removed for other reasons, in which case the observation is censored. In the kidney data set we have the following variables, each one repeated twice for each patient:

- **id**: Unique patient id.
- **time**: Time in (days) from insertion of the catheter to infection in kidney patients using portable dialysis machine.

¹See the Stan website on <https://mc-stan.org/>

- **status**: Censoring indicator (1 if patient is uncensored and 0 otherwise).
- **age**: Age (in years) of each patient.
- **sex**: Sex of each patient (1 if male and 2 if female).
- **disease**: Disease type (GN; AN; PKD; Other).
- **frail**: Frailty estimate from original paper.
- **GN**: Indicator for disease type GN.
- **AN**: Indicator for disease type AN.
- **PKD**: indicator for disease type PKD.

Since every time an individual has had the first recurrence we start counting from 0 until the second recurrence, we say that this data we will be modeling using the *clock-reset* approach. It is, we are interested in the gap time.

In this case study we will model the time (in days) from insertion of a catheter in patient i th to infection by means of the frailty model:

$$\lambda_i(t|\lambda_0, \boldsymbol{\beta}, w_i) = w_i \lambda_0(t) \exp(\beta_2 X_1), \quad i = 1, \dots, n \quad (10)$$

where $X_1 = 1$ if the patient is a female, and 0 if is a male, w_i is the frailty term, β_2 is the regression coefficient for the sex covariate, and n is the number of patients. We will be using the most common frailty distribution $w_i \sim \text{Gamma}(\psi, \psi)$, where $\psi = 1/\kappa$ in terms of the previous notation. As we commented before, in the Bayesian context we should specify a model for the baseline hazard function. In this example we choose the Weibull model, which one of the most widespread models when dealing with survival problems under the Bayesian approach. Then, we know that the baseline hazard has the shape

$$\lambda_0(t) = \lambda \alpha t^{\alpha-1}, \quad (11)$$

where α and $\lambda = \exp(\beta_1)$ are the shape and scale parameters, respectively. Hence, we are assuming $T \sim \text{Weibull}(\alpha, \lambda)$. And this is a convenient parametrization, because if we substitute the specified baseline hazard (11) in the model expression (10), we obtain the new model expression:

$$\lambda_i(t|\lambda_0, \boldsymbol{\beta}, w_i) = w_i \alpha t^{\alpha-1} \exp(\beta_1 + \beta_2 X_1) \quad i = 1, \dots, n.$$

Observe that with the model expression above we have included the parameter related with the Weibull scale parameter into the exponential as an intercept of the regression coefficients.

This will make it easier for us to deal with the parameters. Now, we have to choose a prior distribution for all the parameters playing a role in our model. Taking into account the support of these parameters, we choose some non-informative priors, which are the ones used on a regular basis in the literature. We take $\beta_1, \beta_2 \sim N(0, 1000)$, $\alpha \sim Unif(0, 10)$ and $\psi \sim Gamma(0.01, 0, 01)$.

We have written this frailty model in Stan. To do this, we have taken into account how to deal with censored data. The reader can see the Stan code used in the following listing chunk:

```

1 data {
2   int<lower=0> n;
3   int<lower=0> J;
4   int<lower=0> p;
5   matrix[n,J] time;
6   matrix[n,p] X;
7   matrix[n,J] is_censored;
8 }
9
10 parameters {
11   vector[p] beta;
12   vector<lower=0>[n] w;
13   real<lower=0> psi;
14   real<lower=0> alpha;
15 }
16
17 model {
18   for (i in 1:n) {
19     for (j in 1:J) {
20       if(is_censored[i,j]==1){
21         target += weibull_lccdf(time[i,j] | alpha, (w[i]*exp(beta[1]*X[i,1] + beta[2]*X[i,2]))
22           )^(-1/alpha));
23       } else{
24         time[i,j] ~ weibull(alpha,(w[i]*exp(beta[1]*X[i,1] + beta[2]*X[i,2]))^(-1/alpha));
25       }
26     }
27   }
28   alpha ~ uniform(0, 10); // Shape parameter prior
29   beta ~ normal(0, 1000); // Scale parameter prior
30   w ~ gamma(psi, psi); // Frailty parameter prior
31   psi ~ gamma(0.01,0.01);
32 }
33 generated quantities{
34   real<lower=0> lambda;
35   lambda = exp(beta[1]);
36 }

```

Listing 1: Stan code for the frailty model.

We have run the Stan model for 100000 warm up simulations and for 100000 simulations with a thin parameter of 10, meaning that we keep only one in 10 from the post-warm-up simulations. Once we have executed the Stan model, we have the posterior distributions of the parameters involved in the model in R, and hence we have all we need to do the

inference. Before analyze the results of the Bayesian frailty model, it is important to check the convergence of the MCMC method. This can be done by looking if the chains are well-mixed. We have plotted the chains for the simulations of the parameters β_1 and β_2 , the result is exposed in the Figure 1. As it can be observed the chains are well-mixed, we cannot distinguish between them, meaning that the MCMC has converged. Another metric used to check the convergence is the number of effective parameters and $\hat{R}$, which is the potential scale reduction factor on split chains. The latter quantities also shown the expected values when convergence is reached in our case. These quantities are exposed for all the parameters except for those of the frailties in Table 1.

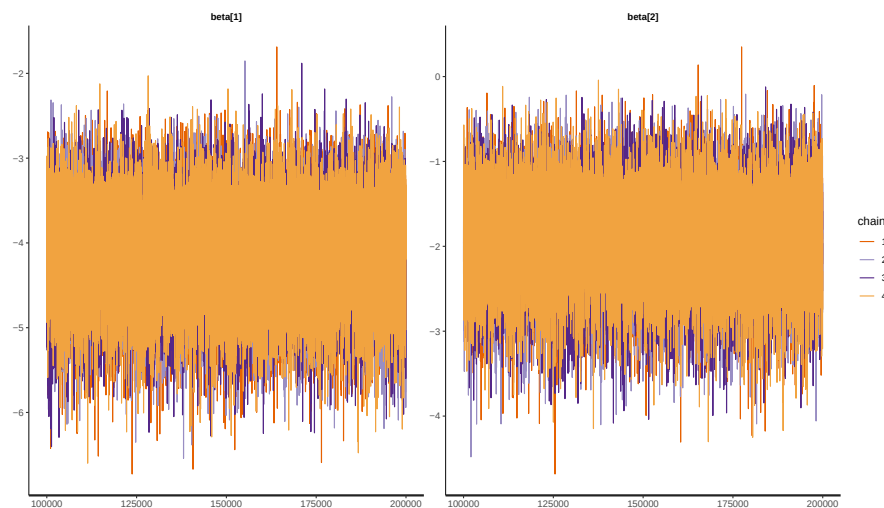


Figure 1: Markov chains trace plots for the simulations of the parameters β_1, β_2 .

Now, it is time to look into the posterior distributions. The reader can observe some summary statistics about the posterior distributions in the Table 1. The results obtained are very similar to the ones presented in [1], this suggest that our Bayesian frailty model is working good.

Parameter	Mean	SD	2.5%	50%	97.5%	n_{eff}	$\hat{R}$
β_1	-4.12	0.59	-5.35	-4.09	-3.02	38838	1
β_2	-1.90	0.56	-3.05	-1.88	-0.87	31455	1
α	1.23	0.16	0.93	1.22	1.57	29240	1
ψ	2.75	5.86	0.79	1.89	8.54	7514	1
λ	0.02	0.01	0.00	0.02	0.05	39166	1

Table 1: Summary statistics of the posterior distributions. In addition the number of effective parameters and the $\hat{R}$ quantities.

In addition, with the aim of compare both models, we have fitted a frequentist frailty model using exactly the model in (10). We have obtained under this approach a point estimate of $\hat{\beta}_2 = -1.38$, which is similar to the one obtained under the Bayesian approach (see in Table 1).

But now, we will exploit the Bayesian advantages when analyzing the results obtained. When working under the frequentist perspective, the results obtained fitting a frailty model are the point estimates of the parameters in the model and its standard error. However, under the Bayesian approach, what we have obtained is the posterior distribution of this parameters. In our case, we have a total amount of 40000 simulations of all the parameters involved in the model. With this simulations one can analyze how are the "estimates".

In the Figure 2 we have the posterior distribution of the regression parameter β_2 , in the red dashed vertical line I have plotted the point estimate of the frequentist approach. As we can see the point estimate of the frequentist model is within the 80% interval of the posterior distribution. Nonetheless, observe that under the Bayesian approach we have more information, because we have the whole distribution of the parameter instead of a point estimate and its standard error. Moreover, the so-called credible intervals (the equivalent to the frequentist confidence interval) have the intuitive meaning: is an interval where the 80% of the probability is allocated, and then the true parameter is within this interval with a 80% probability. In the right part of the Figure 2 we have the posterior distribution of the hazard ratio, it is of $\exp(\beta_2)$.

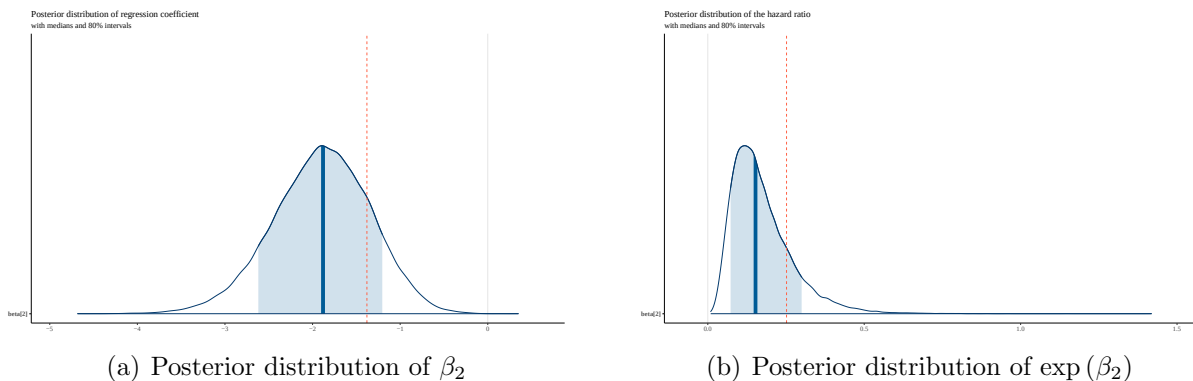


Figure 2: In the left, posterior distribution of β_2 , in the red dashed line the point estimate of the frequentist frailty model. In the right, the same for $\exp(\beta_2)$, the hazard ratio.

We have done the same plots for several frailties of the model, specifically for the frailties of the first four individuals. The plots are exposed in the Figure 3 and Figure 4. As the reader can observe, the frequentist estimate is inside of the 80% credible interval in all the cases.

Inspired by what is done in [1] we have developed the individual survival curve based on pos-

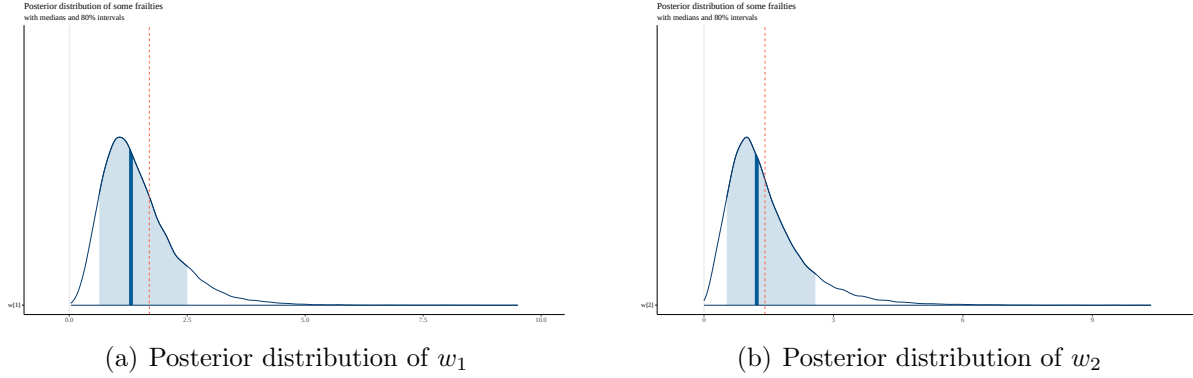


Figure 3: In the left, the posterior distribution of the frailty term for the first individual, where the red dashed line is the frequentist estimation. In the right, the same but for the second individual.

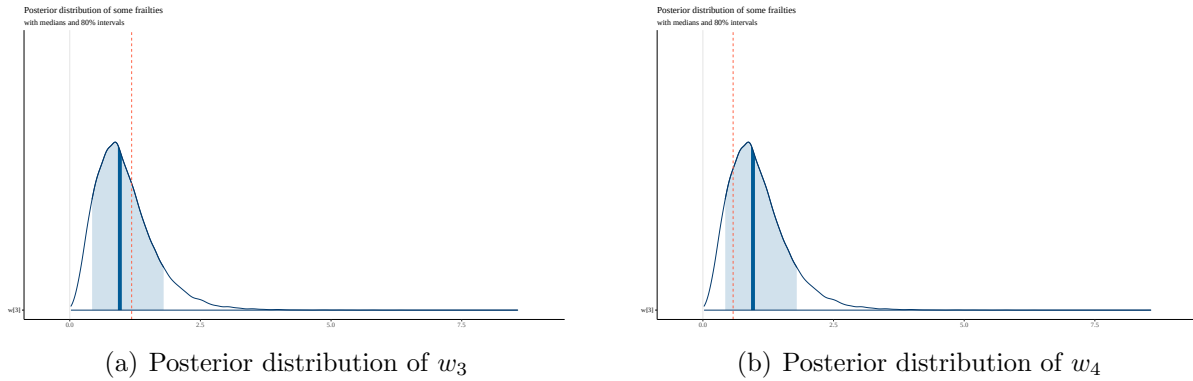


Figure 4: In the left, the posterior distribution of the frailty term for the third individual, where the red dashed line is the frequentist estimation. In the right, the same but for the fourth individual.

terior samples. Since we are working with gap time, the individual survival curve represents the posterior mean of the probability of infection from any catheter insertion at each time considering the two replicates per patient. This curve is exposed in the Figure 5. As we can see the survival curves are higher for females than for males.

In order to make this analysis fully reproducible all the code used has been uploaded to a Github repository. ²

²See my personal GitHub repository on https://github.com/ArnauGF/MSc-UPC/tree/bayesian_survival

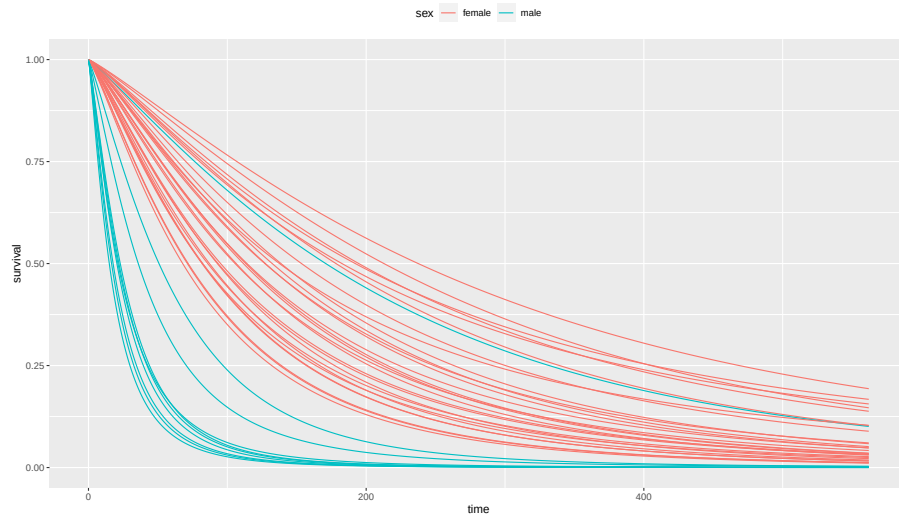


Figure 5: Posterior mean of the individual survival function.

References

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